

04 - Vaginal Progesterone Before Twenty-Four Weeks

Companion reading handout for simultaneous listening.

Course: Medical Prevention of Spontaneous Early Delivery

The cleanest medical-treatment evidence in this course is not progesterone for every patient at risk of preterm birth.

The cleanest evidence is vaginal progesterone for an asymptomatic singleton pregnancy with a sonographic short cervix in the mid-trimester, especially before twenty-four weeks.

That sentence is longer than the usual shorthand, but the length is doing real work. It protects the population, the route, the timing, and the endpoint.

Building Evidence as Clinical Sentences

We will build the evidence as clinical sentences.

Fonseca

Start with the Fonseca trial, published in the New England Journal of Medicine in two thousand seven. The trial began with routine transvaginal cervical-length screening at a median of twenty-two weeks, with a range from twenty to twenty-five weeks. Out of twenty-four thousand six hundred twenty women screened, four hundred thirteen had a cervical length of fifteen millimeters or less. That was one point seven percent.

That is the first memory: Fonseca is a very short-cervix trial. Fifteen millimeters or less. Not twenty-five. Not a broad risk group. A very enriched population.

Two hundred fifty women were randomized to vaginal progesterone or placebo. The regimen was two hundred milligrams vaginal progesterone each night, from twenty-four weeks to thirty-three weeks plus six days, effectively to thirty-four weeks. The primary endpoint was spontaneous delivery before thirty-four weeks.

The result: spontaneous delivery before thirty-four weeks occurred in nineteen point two percent with progesterone and thirty-four point four percent with placebo. The relative risk was zero point five six, with a ninety-five percent confidence interval from zero point three six to zero point eight six.

Now store the trial as a sentence.

Fonseca means: cervix fifteen millimeters or less, vaginal progesterone two hundred milligrams nightly, twenty-four to thirty-four weeks, spontaneous delivery before thirty-four weeks fell from thirty-four point four percent to nineteen point two percent.

That is a memory unit. The name, population, dose, endpoint, and result travel together.

Now move to the PREGNANT trial, the Hassan trial from two thousand eleven.

PREGNANT studied asymptomatic singleton pregnancies with a cervical length between ten and twenty millimeters at nineteen to twenty-three plus six weeks. The regimen was vaginal progesterone gel, ninety milligrams daily. Treatment continued until thirty-six plus six weeks, rupture of membranes, or delivery. The primary endpoint was preterm birth before thirty-three weeks.

The result: preterm birth before thirty-three weeks was eight point nine percent with vaginal progesterone and sixteen point one percent with placebo. The relative risk was zero point five five, with a ninety-five percent confidence interval from zero point three three to zero point nine two.

Store that sentence.

PREGNANT means: asymptomatic singleton pregnancy, cervix ten to twenty millimeters, ninety milligrams vaginal gel daily, before-thirty-three endpoint, risk reduced from sixteen point one percent to eight point nine percent.

Notice how different this is from saying, "Progesterone works." The trial name has become clinically useful because you know where the evidence belongs.

Comparison

Now compare the two trials.

Fonseca used a very short cervix, fifteen millimeters or less, and a two hundred milligram nightly vaginal regimen. PREGNANT used ten to twenty millimeters and a ninety milligram vaginal gel. Fonseca's endpoint was spontaneous delivery before thirty-four weeks. PREGNANT's was preterm birth before thirty-three weeks. Both support vaginal progesterone in the short-cervix phenotype, but neither proves that every patient with a remote history of preterm birth and a normal cervix benefits from vaginal progesterone. Neither proves late initiation at thirty weeks. Neither proves a benefit in active preterm labor.

Meta-Analysis

Meta-analysis then asks whether the pattern holds across trials.

The Romero individual-patient-data meta-analysis from two thousand twelve is the first consolidation to remember. It pooled women with asymptomatic sonographic short cervix, twenty-five millimeters or less, and found reduced preterm birth before thirty-three weeks, with a relative risk of zero point five eight and a confidence interval from zero point four two to zero point eight zero. It also reported neonatal benefits, including lower respiratory distress syndrome and lower composite neonatal morbidity and mortality.

That matters because the course is not just asking whether a birth threshold moves. It is asking whether the fetus is protected from the biology of prematurity. A reduction in a gestational-age endpoint is more convincing when neonatal morbidity moves in the same direction.

The Romero individual-patient-data meta-analysis from two thousand eighteen is especially useful because it reports both binary outcomes and gestational age at delivery. In singleton gestations with a sonographic short cervix, vaginal progesterone reduced preterm birth before thirty-three weeks: thirteen point one percent versus twenty-one point nine percent, relative risk zero point six two. It also increased mean gestational age at delivery by zero point seven four weeks, with a ninety-five percent confidence interval from zero point one eight to one point three weeks.

That zero point seven four weeks deserves careful handling. It is not a huge-sounding number if spoken lazily. But zero point seven four weeks is about five days on average. In obstetrics, a mean shift of several days can matter, especially if part of the distribution is being moved away from the most vulnerable gestational ages. It also teaches that outcomes do not have to be only yes-or-no thresholds. Gestational age at delivery itself can be an outcome.

Guideline Translation

Now add guidelines.

Society for Maternal-Fetal Medicine Consult Series number seventy recommends vaginal progesterone for asymptomatic singleton pregnancies without prior spontaneous preterm birth when the transvaginal cervical length is twenty millimeters or less before twenty-four weeks. For twenty-one to twenty-five millimeters, it recommends considering vaginal progesterone through shared decision-making.

This is a memory gradient.

Twenty-five millimeters names the short-cervix boundary. Twenty millimeters or less is the stronger recommendation zone. Twenty-one to twenty-five millimeters is a discussion zone. That distinction is not pedantry. It is the guideline translating evidence strength into clinical action.

What about OPPTIMUM?

OPPTIMUM is useful because it prevents overconfidence. It studied a broader risk population: previous spontaneous birth at thirty-four weeks or earlier, cervical length twenty-five millimeters or less, positive fetal fibronectin with other risk factors, or other prespecified clinical risks. It used vaginal progesterone two hundred milligrams daily from twenty-two to twenty-four weeks until thirty-four weeks.

Its three primary outcomes were deliberately broader than a single birth threshold: fetal death or delivery before thirty-four weeks, a neonatal composite, and childhood cognitive score at two years. The obstetric primary outcome was eighteen percent with placebo and sixteen percent with progesterone, adjusted odds ratio zero point eight six, with a confidence interval from zero point six one to one point two two. The neonatal primary outcome also did not reach significance, and the two-year cognitive score was essentially unchanged.

That does not erase the short-cervix evidence. It teaches that progesterone efficacy depends on phenotype and endpoint. When the population becomes broader and the endpoint changes, the conclusion can change. A resident should not use OPPTIMUM to dismiss Fonseca or PREGNANT, and should not use Fonseca or PREGNANT to ignore OPPTIMUM. The mature interpretation is narrower and more powerful: vaginal progesterone is best supported when the patient truly belongs to the asymptomatic singleton short-cervix phenotype.

Final Interpretation

A mature resident should therefore carry two truths at once.

First, vaginal progesterone is the best-supported medical treatment for the asymptomatic singleton short cervix detected in the mid-trimester, especially in the strongest evidence window before twenty-four weeks.

Second, that statement is not permission to prescribe progesterone as a universal anti-preterm-birth drug.

If you want a final memory structure for this module, keep three clinical sentences.

Fonseca: fifteen millimeters or less, two hundred milligrams nightly, before-thirty-four fell from thirty-four point four to nineteen point two percent.

PREGNANT: ten to twenty millimeters, ninety milligrams gel daily, before-thirty-three fell from sixteen point one to eight point nine percent.

Romero synthesis: short cervix twenty-five millimeters or less, lower early preterm birth, neonatal morbidity moving in the favorable direction, and in the updated analysis about five mean days gained.

Society for Maternal-Fetal Medicine: asymptomatic singleton, no prior spontaneous preterm birth, before twenty-four weeks, twenty millimeters or less is recommend; twenty-one to twenty-five is discuss.

Those sentences are the antidote to shallow teaching. They make the evidence retrievable because they keep the population, regimen, endpoint, and effect together.